Checkpoint Answer Sheet - Quantized Energy and Atomic Energy Levels

Student: ____	**Date:** ____________________

MLA Standard: MLA.CHEM.ATO.03

Directions: Complete independently after the Lesson Review and Correction page. Use the energy-level information below; do not return to the simulation.

Reference energy levels

Level	n = 1	n = 2	n = 3	n = 4
Relative energy	lowest	higher	higher	highest shown

1. An electron moves from n = 2 to n = 4. Identify the process and explain what must happen to energy.

2. An electron moves from n = 4 to n = 2. Identify the process and explain what leaves the atom.

3. Draw a labeled energy-level model for the n = 4 to n = 2 transition. Include an arrow and label the photon.

Checkpoint Answer Sheet - Spectrum Reasoning

Student: ______________________________________	Date: ___________________

MLA Standard: MLA.CHEM.ATO.03

A hydrogen sample produces four bright visible lines at approximately 410 nm, 434 nm, 486 nm, and 656 nm.

Line	1	2	3	4
Wavelength (nm)	410	434	486	656

4. What feature of this evidence supports the claim that electron energy is quantized?

5. A student says, "The atom can emit any photon energy because electrons can stop between levels." Evaluate and correct the claim.

6. Write a complete claim-evidence-reasoning explanation connecting electron transitions to the observed line spectrum.

Submission check: Save the completed PDF with your name and U2 L6. Upload it to U2 L6 Checkpoint Submission.